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Vision Transformers (*ViT*) and Their Modern Variants in Image Classification

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Introduction / Motivation to the Topic

Vision Transformers (ViT) have recently emerged as one of the most popular and widely cited approaches in computer vision. Inspired by Transformer architectures from natural language processing, ViTs have demonstrated highly competitive performance in image classification tasks, sometimes surpassing traditional convolutional neural networks (CNNs).

This research project aims to investigate ViTs and their modern variants, analyzing their architectures, comparing their performance, and running practical implementations on small datasets in Google Colab. The motivation for selecting this topic is the combination of high academic relevance, strong citation counts, and practical feasibility for a student project.

Topic Description

Vision Transformers (ViT) apply the self-attention mechanism of Transformers to image data. Images are divided into patches, each patch is flattened and embedded, and the sequence of patch embeddings is processed by a standard Transformer encoder. The output class token is then used for classification.

Modern variants of ViT include:

- **DeiT (Data-efficient Image Transformers)** – a lighter, more efficient version of ViT
- **Swin Transformer** – hierarchical structure for better local representation
- **MobileViT** – optimized for mobile and edge devices

These variants allow flexibility in implementation and experimentation, making the topic highly suitable for research and code execution in Google Colab.

Research Goal

Vision Transformers (ViT) apply the self-attention mechanism of Transformers to image data. Images are divided into patches, each patch is flattened and embedded,

and the sequence of patch embeddings is processed by a standard Transformer encoder. The output class token is then used for classification.

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Main Survey Paper

Paper: *Transformers in Vision: A Survey*

Authors: Khan, Naseer, Hayat, Zamir, Shahbaz Khan & Shah

Year: 2021

Citations: ~2,200

Link: <https://arxiv.org/abs/2101.01169>

Description / Justification

This paper is selected as the main survey because it provides a comprehensive overview of Vision Transformers (ViTs) and their applications in computer vision, including:

- Classification
- Detection
- Segmentation
- Video and multimodal tasks

It also discusses architecture, variants, advantages, limitations, and future directions. The paper is highly cited (~2,200 citations), making it academically credible, and is well-structured and readable, which simplifies understanding and summarization.

Supporting Papers

1. *A Survey on Efficient Vision Transformers: Algorithms, Techniques, and Performance Benchmarking* (2023) <https://arxiv.org/abs/2309.02031>
2. *Vision Transformers for Image Classification: A Comparative Survey* (2025) <https://doi.org/10.3390/technologies13010032>
3. *Exploring the Synergies of Hybrid CNNs and ViTs Architectures for Computer Vision: A survey* (2024) <https://arxiv.org/abs/2402.02941>

Introduction

Vision Transformers (ViTs) have emerged as a groundbreaking approach in computer vision by adapting the transformer architecture, originally designed for natural language processing, to image data. Unlike convolutional neural networks (CNNs), which rely on local receptive fields and convolutional kernels, ViTs process images by dividing them into fixed-size patches and using self-attention to model global relationships among patches.

This project aims to explore the structure, variants, and applications of ViTs, and to perform practical experiments using pre-trained models on small datasets in Google Colab. The goal is to provide a comprehensive analysis of ViTs and compare their performance, efficiency, and usability for real-world tasks.

Why ViTs are important:

- *They achieve competitive or superior performance on image classification benchmarks.*
- *They provide a **unified architecture** for multiple vision tasks (classification, detection, segmentation).*
- *They allow **flexible scaling** (tiny models to huge models like ViT-Large).*

Motivation

The topic was chosen due to a combination of **academic relevance, practicality, and research potential**:

1. **High impact:** ViT is widely cited (~2,200 citations), indicating strong influence in the research community.
2. **Colab-friendly experimentation:** Pre-trained models allow hands-on demos without heavy computation.
3. **Comparison opportunity:** Multiple variants exist with different architectures, making it easy to analyze pros, cons, and efficiency.

4. **Pedagogical value:** Easy to explain visually (patch embedding, attention maps) and conceptually simple compared to more complex AI models (GANs, RL).

Background

Transformer Basics

- **Self-Attention:** Calculates a weighted sum of all input tokens, where weights depend on similarity between tokens.
- **Multi-Head Attention:** Captures multiple types of relationships in parallel.
- **Feed-Forward Networks (MLPs):** Process the attention output for each token individually.
- **Positional Encoding:** Since transformers are permutation-invariant, position information must be added to tokens.

Vision Transformers (ViT)

Steps in a ViT model:

1. **Image Patchification:** Split image into fixed-size patches (e.g., 16×16 pixels).
2. **Flattening & Linear Projection:** Each patch is flattened into a vector and projected to a high-dimensional embedding.
3. **Add Positional Encoding:** Ensures the model understands spatial arrangement.
4. **Transformer Encoder:** Several layers of self-attention + MLPs process the patch embeddings.
5. **Classification Head:** Use the special class token for final prediction.

Vision Transformer Variants

<i>Variant</i>	<i>Key Idea</i>	<i>Strengths</i>	<i>Practical Notes</i>
<i>ViT (original)</i>	<i>Standard transformer on patches</i>	<i>Strong on large datasets</i>	<i>Needs pre-trained weights</i>
<i>DeiT</i>	<i>Data-efficient, uses distillation</i>	<i>Can train on smaller datasets</i>	<i>Ideal for Colab</i>
<i>Swin Transformer</i>	<i>Hierarchical, multi-scale</i>	<i>Good for detection/segmentation</i>	<i>Slightly more complex</i>
<i>MobileViT</i>	<i>Lightweight & mobile-friendly</i>	<i>Efficient, small memory</i>	<i>Runs on CPU/low-GPU easily</i>

Practical Implementation

Goal: Show a working example without training from scratch.

Steps:

1. Load pre-trained model using timm.
2. Load small dataset (CIFAR-10 or MNIST).
3. Preprocess images to 224×224 and normalize.
4. Pass through the model and observe outputs.

Example Python snippet:

```
!pip install timm
import timm, torch
from torchvision import datasets, transforms

model = timm.create_model('vit_tiny_patch16_224', pretrained=True)
model.eval()

transform = transforms.Compose([transforms.Resize((224,224)),
transforms.ToTensor()])
```



```
dataset = datasets.CIFAR10(root='./data', train=False, download=True,  
transform=transform)  
image, label = dataset[0]  
image = image.unsqueeze(0)  
output = model(image)  
print("Output logits:", output)
```

Comparison and Analysis

Metrics to compare ViT variants:

- Accuracy on benchmark datasets (ImageNet, CIFAR-10)
- Number of parameters
- Inference speed / latency
- Resource requirement (CPU/GPU)
- Suitability for deployment (mobile, edge, desktop)

Initial Observations (from surveys):

- DeiT is ideal for small-scale experiments (low compute, high accuracy).
- Swin Transformer improves performance for detection/segmentation.
- MobileViT balances accuracy with efficiency.
- Original ViT requires large datasets and is heavy for small-scale demos.